⚽ FIFA Player Performance Analysis

📌 Overview

This project analyzes FIFA player performance and market value data using PostgreSQL to uncover meaningful insights related to player performance, club investments, transfer risks, and talent identification.

Modern football clubs rely heavily on data-driven decisions for player recruitment, contract negotiations, injury management, and squad planning. The goal of this project was to simulate real-world sports analytics scenarios and demonstrate practical SQL skills by solving business problems using data.

The project includes:

- Loading and exploring the dataset
- Performing Exploratory Data Analysis (EDA)
- Cleaning and validating data
- Running analytical SQL queries in PostgreSQL
- Creating a detailed project report
- Designing a presentation using Gamma AI

📊 Dataset

The dataset contains information about football players across different clubs, positions, and nationalities.

Dataset Statistics

Metric	Value
Total Players	2800
Total Clubs	7
Total Nationalities	8
Total Positions	9
Average Age	28 Years
Average Overall Rating	76.87
Average Market Value	€90.57 Million
Injury Prone Players	24.1%

Dataset Features

- Player ID
- Player Name
- Age
- Nationality
- Club
- Position
- Overall Rating
- Potential Rating
- Matches Played
- Goals
- Assists
- Minutes Played
- Market Value
- Contract Years Left
- Injury Prone Status
- Transfer Risk Level

The dataset was used to analyze player performance, transfer decisions, and club strategies.

🛠 Tools and Technologies

- **PostgreSQL** – Database management and analysis
- **SQL** – Data querying and business analysis
- **Gamma AI** – Presentation creation and storytelling
- **Git & GitHub** – Version control and project hosting

🔍 Project Workflow

1. Data Loading

- Imported the dataset into PostgreSQL.
- Created the required database and tables.

2. Exploratory Data Analysis (EDA)

- Explored dataset structure.
- Checked total players, clubs, and nationalities.

- Analyzed distributions across positions and clubs.

3. Data Cleaning

- Checked for missing values.
- Validated column consistency.
- Verified data quality before analysis.

4. SQL Analysis

- Performed SQL queries to answer real-world business questions.
- Applied SQL concepts including Basic to Advanced including Joins, Subqueries, CTEs, Window Functions, Aggregate Functions and CASE statements.

5. Reporting

- Prepared a detailed project report summarizing findings and recommendations.

6. Presentation Creation

- Created a professional project presentation using Gamma AI to communicate insights effectively.

🏗 Project Architecture

Dataset CSV

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PostgreSQL Database

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Data Cleaning

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EDA

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SQL Analysis

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Business Insights

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Report Generation

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Presentation (Gamma AI)

💾 SQL Concepts Implemented

1. Database and Table Creation

- Created database and table structure.
- Applied primary key constraints.
- Used appropriate data types.

2. Data Exploration

- Displayed sample records.
- Identified missing values.
- Explored distinct nationalities.
- Counted total players.

3. Basic SQL Analysis

- Average player market value.
- Top 10 rated players.
- Players older than 30.
- Player count by position.
- Highest average rated club.
- Country with the most players.

4. Player Performance Analysis

- Most valuable young players.
- Players with high potential growth.
- Goal contribution leaders.
- Goals per 90 minutes analysis.
- Best players by position.

5. Transfer Market Analysis

- Overvalued players.
- Undervalued players.
- Transfer risk players with expiring contracts.
- Market value comparisons with club averages.

6. Injury Analysis

- High-value injury-prone players.

- Clubs investing heavily in injury-prone players.
- Clubs with the highest injury counts.

7. Club Level Analytics

- Total goals scored by each club.
- Average rating by position.
- Total club market value.

8. SQL Concepts

Category	Concepts
Basic SQL	SELECT, WHERE, GROUP BY, ORDER BY
Intermediate SQL	CASE Statements, Aggregate Functions, Subqueries
Advanced SQL	Joins, CTEs, Window Functions

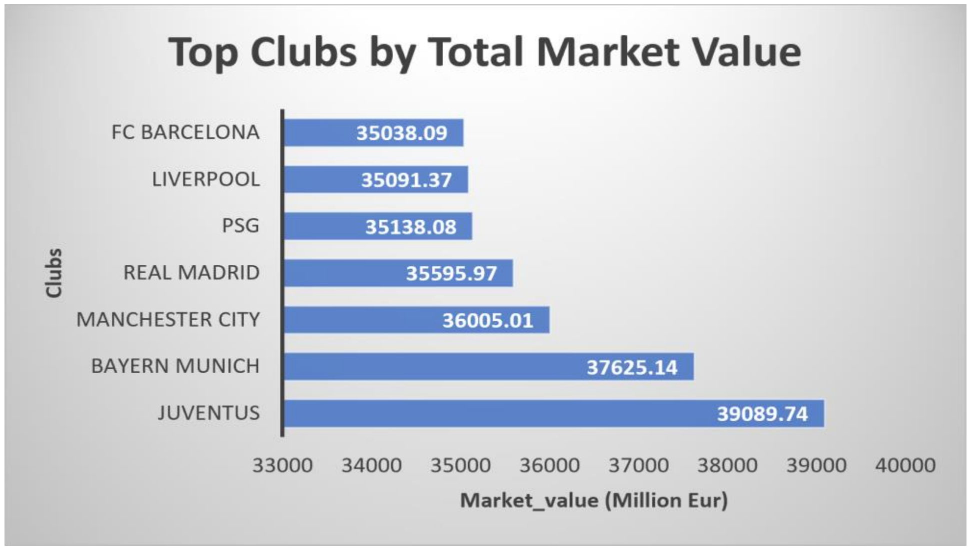
❓ Business Questions

- Create a report showing:
  - `player name`
  - `overall rating`
  - `market value million eur`
  - `age`
  - `player category`
  - `age group`
  - `value category` using three **CASE statements** in a single query?
- Find clubs where total market value exceeds €30,000 million.
- Which players are more valuable than the average player in the club?
- Which player is the most expensive in each playing position?
- Find the players with the largest gap between potential and overall rating?
- Which clubs have an average market value greater than €90 million?
- Which clubs spend more than €200 million on injury-prone players?
- Find the second highest-rated player in each club?
- Find players whose market value is higher than the previous player in their club?
- Find the top scorer and second top scorer in each club?

📊 Data Visualizations and Insights

1. Title: 📈 Top Clubs by Total Market Value (€ Million)

Chart Image:

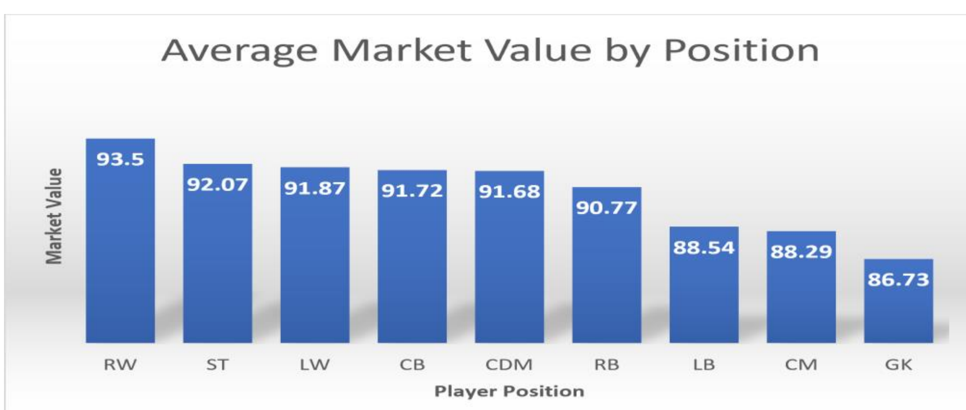


Key Insights:

- Juventus recorded the highest squad market value at €39.09 billion.
- FC Barcelona recorded the lowest squad market value €35.09 billion.
- The difference between the highest and lowest club valuation was €4B.

2. Title: 💰 Average Market Value Across Playing Positions

Chart Image:

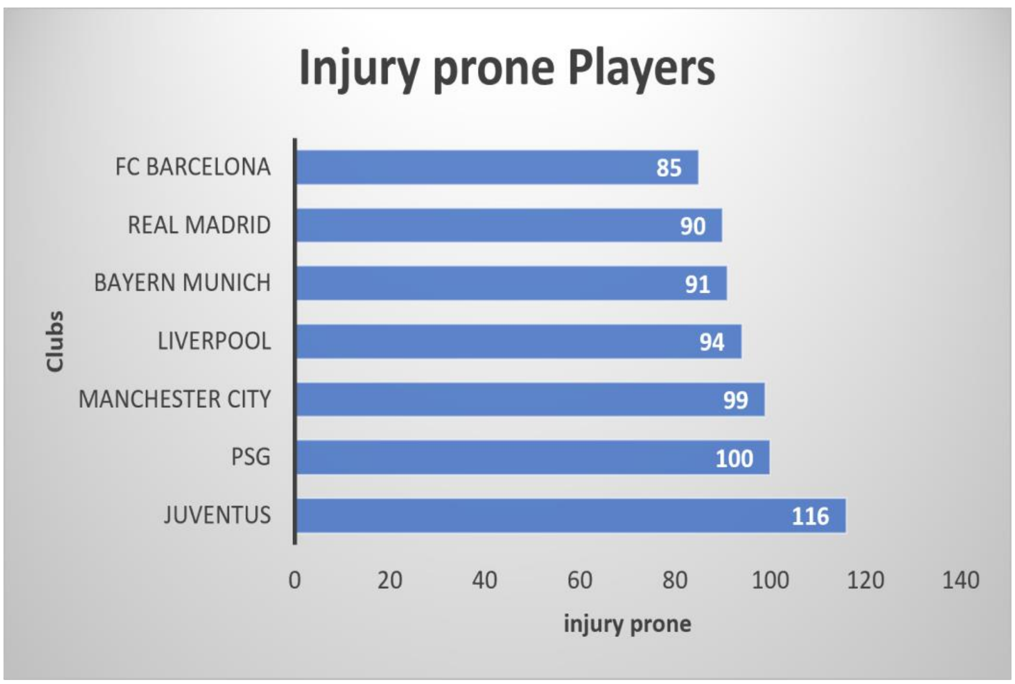


Key Insights:

- RW players have the highest average market value (**€93.50M**).
- GK players have the lowest average market value (**€86.73M**).
- Attacking positions command higher transfer valuations than defensive roles.
- The market value gap between positions is relatively small (**€6.77M**).
- Clubs maintain balanced investments across different positions.

3. Title: 🚑 Distribution of Injury-Prone Players Across Clubs

Chart Image:

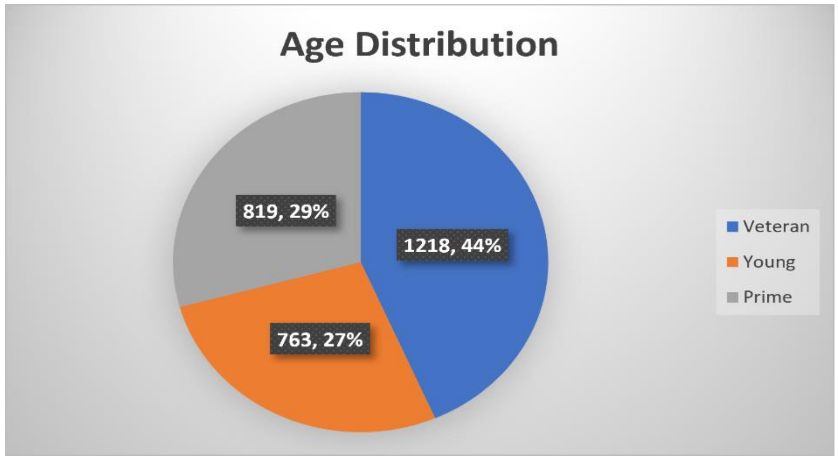


Key Insights:

- Juventus recorded the highest number of injury-prone players (116).
- PSG and Manchester City also showed significant injury exposure.
- FC Barcelona had the lowest injury-prone player count (85).
- Higher injury counts can increase financial and squad management risks.

4.Title: 👥 Player Age Group Distribution

Chart Image:



Key Insights:

• Veterans dominate the dataset with **44%** of players (1218 players).

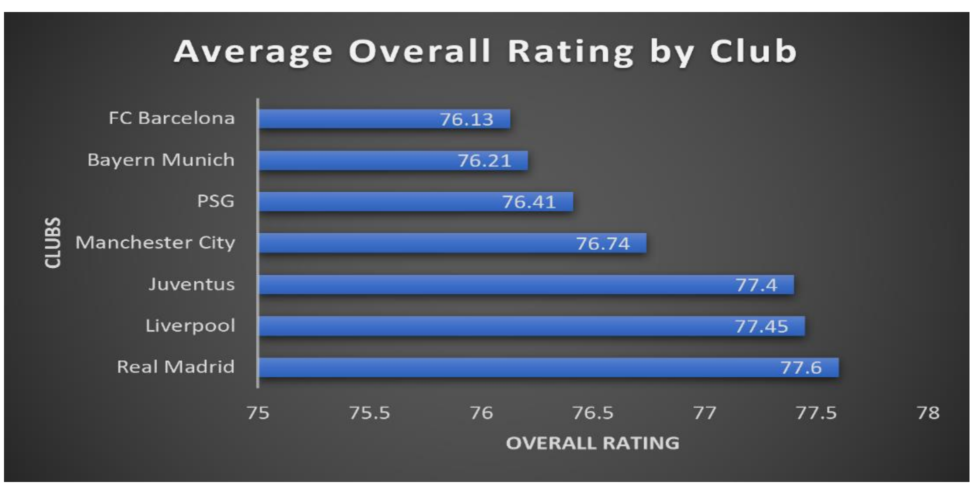
• Young players account for **27%**, representing future talent (763 players).

• Prime-age players make up **29%**, ensuring competitive performance. (819 players).

• Balanced age profiles support long-term squad stability.

5.Title: 🏆 Average Player Rating by Club

Chart Image:



Key Insights:

• Real Madrid led with an average rating of **77.60**.

• Liverpool and Juventus followed closely with **77.45** and **77.40**.

• Club ratings varied by only **1.47 points**, indicating strong competition.

• Every club maintained an average rating above **76**

💻 Sample SQL Queries

1.Which players are Elite, Good, Average, or Poor, group players by transfer value and categorize players by age as young, prime, and veteran?

```
SELECT player_name, overall_rating, market_value_million_eur, age,
    CASE
        WHEN overall_rating > = 90 THEN 'Elite'
        WHEN overall_rating BETWEEN 80 AND 89 THEN 'Good'
        WHEN overall_rating BETWEEN 70 AND 79 THEN 'Average'
        ELSE 'Poor'
        END AS player_category,
    CASE
        WHEN market_value_million_eur >= 150 THEN 'Superstar'
        WHEN market_value_million_eur BETWEEN 100 AND 149.99 THEN
         'World Class'
        WHEN market_value_million_eur BETWEEN 50 AND 99.99 THEN
          'Valuable'
        ELSE 'Budget Player'
        END AS value_category,
    CASE
        WHEN age < 23 THEN 'Young'
```

```sql
            WHEN age BETWEEN 23 AND 29 THEN 'Prime'
            ELSE 'Veteran'
            END AS age_group
FROM player_analysis;
```

2.Which player is the most expensive in each playing position?

```sql
    SELECT player_position, player_name, market_value_million_eur
      FROM player_analysis p1
      WHERE market_value_million_eur = (
                              SELECT MAX(market_value_million_eur)
                                FROM player_analysis p2
                                WHERE p1.player_position = p2.player_position
);
```

3.Find the players with the largest gap between potential and overall rating?

```sql
WITH player_gap AS (
      SELECT player_name, potential_rating, overall_rating,
           (potential_rating - overall_rating) AS improvement_gap
        FROM player_analysis
)
      SELECT player_name, improvement_gap
         FROM player_gap
         WHERE improvement_gap = (
                            SELECT MAX(improvement_gap) AS highest_gap
                                FROM player_gap
);
```

4.Find the top scorer and second top scorer in each club?

```sql
WITH scorer_rank AS (
    SELECT player_name, club, goals,
```

```
ROW_NUMBER() OVER(PARTITION BY club ORDER BY goals DESC) AS
    goals_rank
    FROM player_analysis
)
SELECT player_name, club, goals, goals_rank
FROM scorer_rank
WHERE goals_rank IN (1, 2)
ORDER BY club, goals_rank;
```

5.Find clubs where total market value exceeds €30000 million

```
SELECT club, SUM(market_value_million_eur) AS total_market_value
FROM player_analysis
GROUP BY club
HAVING SUM(market_value_million_eur) > 30000
ORDER BY total_market_value DESC;
```

📊 Key Insights

- 📈 Young players show the highest improvement potential.
- 💰 Several high-performing players are undervalued in the transfer market.
- ⚠️ Injury-prone players represent significant financial risk for clubs.
- 📄 Players nearing contract expiration increase transfer risk.
- 🎯 Goals per 90 minutes reveal hidden high-impact performers.

💡 Business Recommendations

🔍 1. Prioritize scouting high-potential young players

- Focus recruitment efforts on players with large improvement gaps.
- Sign them before their market value rises significantly.

💰 2. Target undervalued players in the transfer market

- Use data-driven scouting to identify players delivering high performance at lower costs.
- This improves squad quality while controlling spending.

⚠️ 3. Reduce investment in injury-prone players

- Include injury history as a major factor during transfer decisions.
- Avoid allocating a large portion of transfer budgets to high-risk players.

📄 4. Renew contracts for key players early

- Players with less than one year remaining should be prioritized for renewal.
- This protects club assets and preserves transfer value.

📈 5. Use performance efficiency metrics in player evaluation

- Consider goals per 90 minutes and assists per 90 minutes instead of only total goals.
- This provides a fair comparison between starters and substitutes.

🏆 6. Build balanced squads instead of relying on star players

- Teams dependent on one or two players become vulnerable to injuries or transfers.
- Developing depth improves long-term competitiveness.

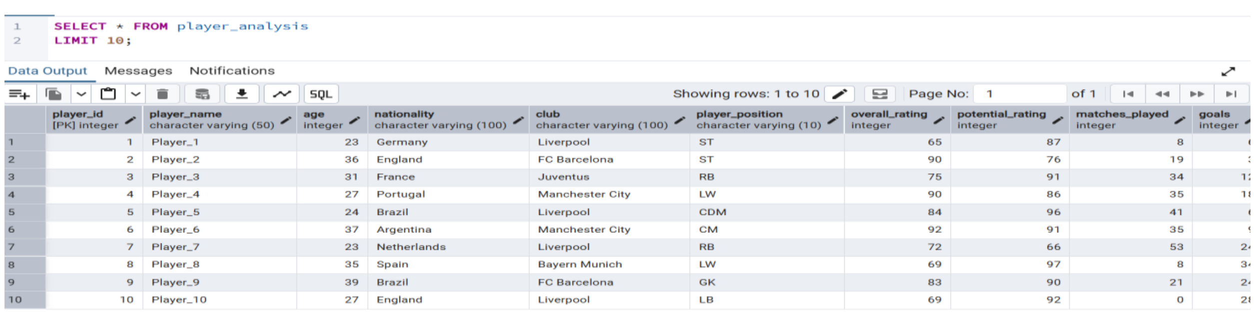
📊 7. Use analytics for transfer decisions

- Combine ratings, market value, age, potential, and injury history to evaluate transfers.
- Data-driven decisions reduce transfer risk and improve return on investment.

📷 Project Screenshots

1.Dataset Preview:

The fifa_player_performance_analysis.CSV dataset imported into PostgreSQL for analysis.



```sql
SELECT * FROM player_analysis
LIMIT 10;
```

Data Output Messages Notifications

Showing rows: 1 to 10 Page No: 1 of 1

	player_id [PK] integer	player_name character varying (50)	age integer	nationality character varying (100)	club character varying (100)	player_position character varying (10)	overall_rating integer	potential_rating integer	matches_played integer	goals integer
1	1	Player_1	23	Germany	Liverpool	ST	65	87	8	[illegible]
2	2	Player_2	36	England	FC Barcelona	ST	90	76	19	[illegible]
3	3	Player_3	31	France	Juventus	RB	75	91	34	[illegible]
4	4	Player_4	27	Portugal	Manchester City	LW	90	86	35	[illegible]
5	5	Player_5	24	Brazil	Liverpool	CDM	84	96	41	[illegible]
6	6	Player_6	37	Argentina	Manchester City	CM	92	91	35	[illegible]
7	7	Player_7	23	Netherlands	Liverpool	RB	72	66	53	[illegible]
8	8	Player_8	35	Spain	Bayern Munich	LW	69	97	8	[illegible]
9	9	Player_9	39	Brazil	FC Barcelona	GK	83	90	21	[illegible]
10	10	Player_10	27	England	Liverpool	LB	69	92	0	[illegible]

2.Database Schema:

Structure of the ‘player_analysis’ table containing players attributes used for analysis.

Query Query History

```
SELECT column_name, data_type
FROM information_schema.columns
WHERE table_name = 'player_analysis';
```

Data Output Messages Notifications

SQL

	column_name name	data_type character varying
1	player_id	integer
2	matches_played	integer
3	goals	integer
4	assists	integer
5	minutes_played	integer
6	market_value_million_...	numeric
7	contract_years_left	integer
8	age	integer
9	overall_rating	integer
10	potential_rating	integer
11	player_name	character varying
12	transfer_risk_level	character varying
13	nationality	character varying
14	club	character varying
15	player_position	character varying
16	injury_prone	character

3.SQL Query Execution and Results:

1.Which players are Elite, Good, Average, or Poor, group players by transfer value and categorize players by age as young, prime, and veteran?

Query Query History

```
SELECT player_name, overall_rating, market_value_million_eur, age,
            CASE
                WHEN overall_rating >= 90 THEN 'Elite'
                WHEN overall_rating BETWEEN 80 AND 89 THEN 'Good'
                WHEN overall_rating BETWEEN 70 AND 79 THEN 'Average'
                ELSE 'Poor'
                END AS player_category,
            CASE
                WHEN market_value_million_eur >= 150 THEN 'Superstar'
                WHEN market_value_million_eur BETWEEN 100 AND 149.99 THEN 'World Class'
                WHEN market_value_million_eur BETWEEN 50 AND 99.99 THEN 'Valuable'
                ELSE 'Budget Player'
                END AS value_category,
            CASE
                WHEN age < 23 THEN 'Young'
                WHEN age BETWEEN 23 AND 29 THEN 'Prime'
                ELSE 'Veteran'
                END AS age_group
FROM player_analysis;
```

Data Output Messages Notifications

SQL Showing rows: 1 to 1000

	player_name character varying (50)	overall_rating integer	market_value_million_eur numeric (10,2)	age integer	player_category text	value_category text	age_group text
1	Player_1	65	122.51	23	Poor	World Class	Prime
2	Player_2	90	88.47	36	Elite	Valuable	Veteran
3	Player_3	75	20.24	31	Average	Budget Player	Veteran
4	Player_4	90	164.29	27	Elite	Superstar	Prime
5	Player_5	84	121.24	24	Good	World Class	Prime

2. Which player is the most expensive in each playing position?

Query   Query History

```sql
SELECT player_position, player_name, market_value_million_eur
    FROM player_analysis p1
    WHERE market_value_million_eur = (
                                        SELECT MAX(market_value_million_eur)
                                            FROM player_analysis p2
                                            WHERE p1.player_position = p2.player_position
);
```

Data Output   Messages   Notifications

SQL

player_position character varying (10)	player_name character varying (50)	market_value_million_eur numeric (10,2)
CDM	Player_320	179.81
ST	Player_573	179.70
CM	Player_687	178.93
RB	Player_1297	179.78
RW	Player_1714	178.86
GK	Player_1743	179.27
CB	Player_2147	178.69
LW	Player_2703	179.81
LB	Player_2794	179.96

3.Find clubs where total market value is greater than €20,000 million?

Query   Query History

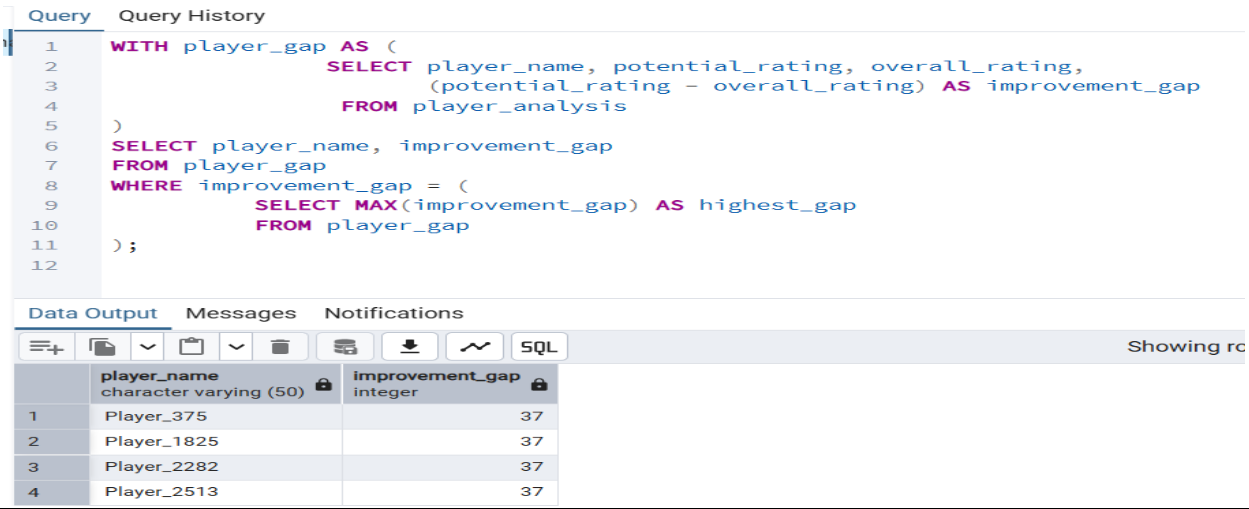
```sql
SELECT club, SUM(market_value_million_eur) AS total_market_value
 FROM player_analysis
 GROUP BY club
 HAVING SUM(market_value_million_eur) > 20000
 ORDER BY total_market_value DESC;
```

Data Output   Messages   Notifications

SQL

	club character varying (100)	total_market_value numeric
1	Juventus	39089.74
2	Bayern Munich	37625.14
3	Manchester City	36005.01
4	Real Madrid	35595.97
5	PSG	35138.08
6	Liverpool	35091.37
7	FC Barcelona	35038.09

4.Find players with the largest gap between potential and overall rating



Query | Query History

```sql
WITH player_gap AS (
                    SELECT player_name, potential_rating, overall_rating,
                           (potential_rating - overall_rating) AS improvement_gap
                     FROM player_analysis
)
SELECT player_name, improvement_gap
FROM player_gap
WHERE improvement_gap = (
          SELECT MAX(improvement_gap) AS highest_gap
          FROM player_gap
);
```

Data Output | Messages | Notifications

SQL | Showing rc

	player_name character varying (50)	improvement_gap integer
1	Player_375	37
2	Player_1825	37
3	Player_2282	37
4	Player_2513	37

🚀 How to Run This Project

Step 1:

Clone the repository:

```
git clone https://github.com/rahul36-glitch/fifa-player-performance-analysis-sql.git
```

Step 2:

Open any relational database like (PostgreSQL, MySQL, SSMS) and create a new database. In this project I used PostgreSQL.

Step 3:

Import the dataset into PostgreSQL.

Step 4:

Execute the SQL script provided in the repository.

Step 5

Review the generated insights, report, and presentation.

📂 Project Structure

```
fifa-player-performance-analysis-sql/
│
├── README.md
├── dataset/
│   └── fifa_player_performance_analysis.csv
├── sql_queries/
│   └── player_analysis.sql
│     ├──data_cleaning.sql
│     ├──eda.sql
│     ├──business_questions.sql
│     ├──advanced_queries.sql│
├── report/
    └── FIFA_Player_Performance_Analysis_Report.pdf
    └── FIFA_Player_Performance_Analysis_Presentation.pdf
```

🎯 Skills Demonstrated

- SQL
- PostgreSQL
- Data Cleaning
- Exploratory Data Analysis (EDA)
- Window Functions
- CTEs
- Subqueries
- Business Analysis
- Reporting and Presentation
- Data Storytelling

🚀 Future Improvements

- 🐍 Automate data cleaning and preprocessing using Python and Pandas.
- 📊 Develop an interactive Power BI dashboard to visualize player performance, club statistics, and transfer market trends.
- 🌍 Expand the dataset by including multiple FIFA editions and league-specific statistics for deeper analysis.
- 🤖 Build predictive models to estimate player market value and future performance using machine learning techniques.
- ⚙️ Automate reporting and insight generation workflows.
- 🎯 Develop a data-driven player scouting recommendation system.
- 🔚 Automate the ETL pipeline using Python and PostgreSQL.

📝 Conclusions

- This project analyzed FIFA player data using SQL and PostgreSQL to identify patterns in player performance, market value, potential growth, and club strategies. Through data cleaning, exploratory analysis, and advanced SQL queries, several valuable insights were generated, including identifying high-potential young players, undervalued transfer opportunities, injury-related financial risks, and player efficiency metrics.
- The analysis demonstrates how data-driven decision-making can support clubs in scouting, transfer planning, contract management, and squad development. It also highlights the practical application of SQL in solving real-world business problems and extracting actionable insights from large datasets.
- Overall, this project strengthened skills in SQL, PostgreSQL, data analysis, and business intelligence while showcasing the ability to convert raw data into meaningful recommendations for stakeholders.

📫 Contact

If you have any feedback, suggestions, or collaboration opportunities, feel free to connect with me.

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- GitHub: https://github.com/rahul36-glitch
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I am always open to discussions about data analytics, SQL, and entry-level and Internship opportunities in the data domain